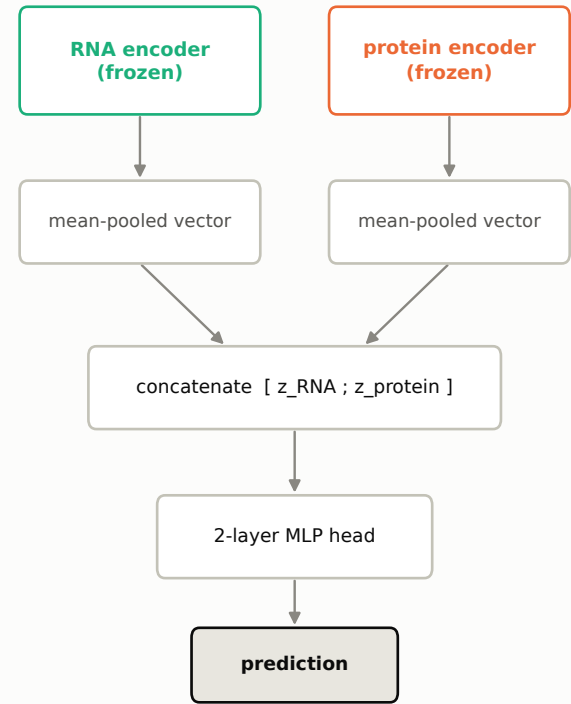
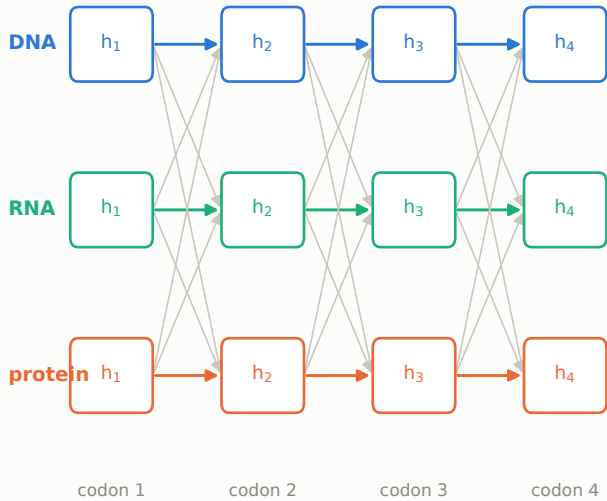


**a Concatenation + MLP (reference baseline B4)    b Coupled Mamba on the codon grid (priority 1)    c Cross-attention, IsoFormer style (priority 2)**

*no token interaction; sets the bar every fusion model must clear*



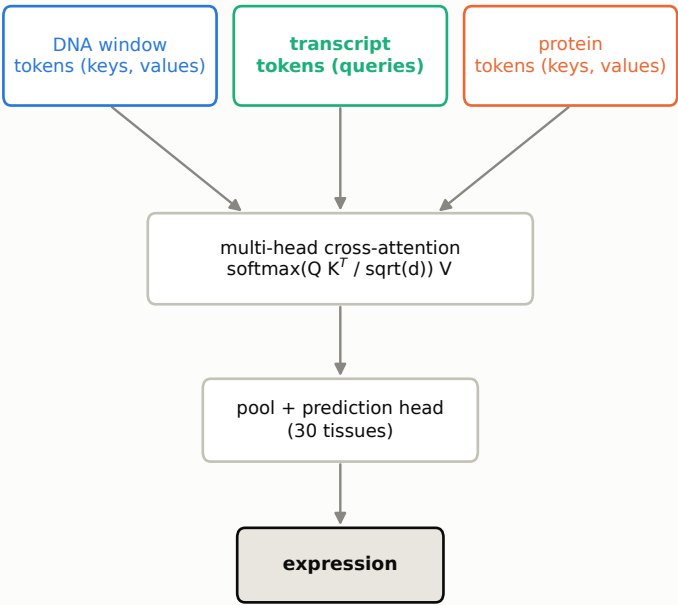
*one recurrent chain per modality, coupled at every codon step  $t$*



$$h_t^m = S_m \text{ (sum over modalities of } h_{t-1}) + B_m x_t^m$$

*linear in sequence length; no alignment loss;  
needs the shared codon grid*

*token streams keep their own lengths; one stream queries the others*



*quadratic in length; no shared grid needed;  
also the benchmark competitor on stability*